

Simulation of the Claus Process for Sulfur Recovery

A Three-Stage Equilibrium-Based Process Simulation Using Aspen Plus V14

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Abstract

Hydrogen sulfide, a toxic byproduct of natural gas processing and petroleum refining, cannot be vented or flared directly without severe environmental consequence. The Claus process remains the standard industrial method for converting H_2S into elemental sulfur — a commodity product with direct market value. This report presents a steady-state simulation of a three-stage Claus unit built in Aspen Plus V14, using the SR-POLAR property package to account for the non-ideal phase behavior of polar and associating species in the gas mixture.

The simulation models a reaction furnace (RGibbs), three catalytic converters (REquil), three interstage condensers (Flash2), and three reheaters (Heater). Feed to the unit consists of an acid gas stream at 171.1 mol/s containing 36.04 mol% H_2S and 53.16 mol% CO_2 , combusted with air at 181.5 mol/s and supplemented by a hydrocarbon-rich fuel gas at 3.25 mol/s. The reaction furnace reaches an adiabatic equilibrium temperature of 995.6°C, after which process gas passes through successive reheating and conversion stages at 260°C, 220°C, and 190°C. All condensers operate at 130°C to maximize liquid sulfur recovery while remaining above the sulfur dew point.

The simulation converges to an overall H_2S -to-sulfur recovery efficiency of 98.70%, with liquid sulfur production across condensers C, C1, and C2 totaling approximately 6,023 kg/hr as combined S_2 and S_8 species. Residual H_2S in the tail gas (OFFGAS stream) measures 2.893 kmol/hr, consistent with the thermodynamic limitations of three-stage Claus operation. Results are discussed in the context of equilibrium constant trends, inter-stage conversion distribution, and sulfur species speciation across the unit.

1. Introduction

Sulfur enters hydrocarbon feedstocks as organic sulfides, mercaptans, and, most abundantly, hydrogen sulfide dissolved in sour natural gas and crude oil. Amine absorption units — typically using MEA, DEA, or MDEA as solvents — strip this H_2S from the gas phase and regenerate a concentrated acid gas stream. That stream cannot simply be incinerated: uncontrolled SO_2 emissions from flaring are regulated under environmental legislation in virtually every country with a refining industry. The Claus process, commercialized in its modern two-stage catalytic form by I.G. Farben in the 1930s and later refined by the oil industry through the mid-twentieth century, solves this by converting the sulfur to its elemental form — a stable, low-toxicity solid that is handled, stored, and sold as a commodity feedstock for sulfuric acid and fertilizer manufacture.

The chemistry operates in two distinct zones. In the thermal stage, roughly one-third of the H_2S feed is combusted with a controlled sub-stoichiometric supply of air to produce SO_2 , with the $\text{H}_2\text{S}:\text{SO}_2$

molar ratio in the furnace outlet controlled to 2:1 — the stoichiometric requirement for the downstream catalytic reaction. This combustion occurs at temperatures between 900°C and 1400°C, producing an equilibrium mixture rich in elemental sulfur vapors (primarily S₂ and S₈) alongside unreacted H₂S, SO₂, CO₂, and water. The exothermic heat of combustion is removed in a waste-heat boiler, then the cooled gas passes through the first condenser, where sulfur condenses and is drained. The remaining vapor — still carrying substantial H₂S and SO₂ — is reheated and passed over an activated alumina or titanium dioxide catalyst where the Claus reaction proceeds:



Because this reaction is exothermic and equilibrium-limited, lower temperatures favor higher conversions — which is why each successive converter operates at a progressively lower inlet temperature. Three catalytic stages, as simulated here, are standard practice for units targeting recovery in the 96–99% range before tail gas treatment. Beyond three stages, the marginal gain in sulfur recovery does not justify the capital cost.

Process simulation has become indispensable in the design and troubleshooting of Claus units. Equilibrium-based models, while they do not capture catalyst deactivation or kinetic limitations in aged beds, provide accurate predictions of thermodynamic performance limits and are widely used for design basis calculations, sensitivity studies, and regulatory compliance reporting. Aspen Plus, with its comprehensive thermodynamic property packages and flexible block architecture, is the standard commercial tool for this purpose.

This report documents the construction, validation, and analysis of a steady-state three-stage Claus simulation. The work is part of a broader portfolio of process simulation projects that also includes a crude distillation unit optimization study and a DEA-based natural gas sweetening simulation — the latter providing the acid gas feed characterization context for this unit. The specific objectives are: (1) to construct a thermodynamically rigorous flowsheet representing industrial Claus operation; (2) to analyze inter-stage conversion behavior and equilibrium constant trends; and (3) to quantify overall sulfur recovery against the theoretical three-stage limit.

2. Feed Characterization

2.1 Stream Compositions and Inlet Conditions

Three feed streams enter the Claus unit: the acid gas from amine regeneration, combustion air, and a supplementary fuel gas used to stabilize the reaction furnace flame. Their compositions and inlet conditions are given in Table 1

Parameter	Acid Gas	Air	Fuel Gas
Pressure (bar)	1.77	1.68	6.00
Temperature (°C)	218	220	40
Flow rate (mol/s)	171.1	181.5	3.25
CO ₂ (mol fr.)	0.5316	—	0.0103
N ₂ (mol fr.)	—	0.7300	0.0368
CH ₄ (mol fr.)	0.0090	—	0.8946
C ₂ H ₆ (mol fr.)	—	—	0.0566
C ₃ H ₈ (mol fr.)	—	—	0.0017
H ₂ S (mol fr.)	0.3604	—	—
O ₂ (mol fr.)	—	0.1950	—
H ₂ O (mol fr.)	0.0990	0.0750	—

Table 1. Feed stream compositions and inlet conditions.

2.2 Acid Gas

The acid gas stream carries 36.04 mol% H₂S, 53.16 mol% CO₂, and 9.90 mol% H₂O, with a small methane fraction (0.90 mol%). This composition is characteristic of a lean amine system treating a gas with a relatively high CO₂:H₂S ratio — a condition that bears directly on furnace design. The CO₂ content is high enough that a portion reacts in the furnace to form COS and CS₂ via side reactions with H₂S and sulfur vapors. Both species are confirmed in the simulation output: COS at 3.863 kmol/hr and CS₂ at 6.44×10⁻³ kmol/hr in the hot gas leaving the furnace. While these quantities are minor relative to the sulfur mass balance, they represent incomplete sulfur recovery pathways and, in real plant operation, would be addressed by catalyst selection (TiO₂ for COS and CS₂ hydrolysis) in the first catalytic bed.

The stream enters at 218°C and 1.77 bar. The elevated temperature reflects the preheat applied after amine regeneration to prevent condensation in the feed piping, particularly important given the high water content. In Aspen Plus, this stream is modeled as a vapor-phase mixture — phase equilibrium confirms it is fully gaseous at these conditions.

2.3 Combustion Air

Combustion air enters at 220°C and 1.68 bar, preheated to avoid quenching the furnace flame. The dry composition is 73.00 mol% N₂ and 19.50 mol% O₂ with 7.50 mol% moisture, giving an O₂:N₂ ratio consistent with atmospheric air. At a flowrate of 181.5 mol/s, the air:acid gas molar ratio is 1.063, which the RGibbs furnace block optimizes to achieve near-complete O₂ consumption while maintaining the 2:1 H₂S:SO₂ ratio in the furnace outlet. The simulation confirms O₂ is effectively fully consumed in the furnace product (residual O₂ = 9.23×10⁻¹² kmol/hr), which validates that the air ratio is correctly specified.

2.4 Fuel Gas

The supplementary fuel gas is predominantly methane (89.46 mol%) with ethane (5.66 mol%), propane (0.17 mol%), and traces of CO₂ and N₂. It enters cold at 40°C and 6 bar, the higher-pressure reflecting pipeline supply conditions — at a relatively small flowrate of 3.25 mol/s. The function of this stream is to raise and stabilize the furnace adiabatic flame temperature, since very lean acid gases (low H₂S content) burn poorly and may not sustain a stable flame without supplemental fuel. In the simulation, complete combustion of the hydrocarbons is captured by the RGibbs block at chemical equilibrium: C₂H₆ and C₃H₈ are entirely absent from the furnace product stream, and CH₄ traces at 3.74×10⁻⁸ kmol/hr — negligible. The carbon from hydrocarbon combustion appears primarily as CO₂ (316.67 kmol/hr in HOT-GAS) and as CO (24.42 kmol/hr), the latter formed by incomplete oxidation under the sub-stoichiometric air conditions necessary for H₂S:SO₂ ratio control.

2.5 Mass Balance Overview

The overall flowsheet mass balance, summarized in Table 2, closes to within 2.94×10⁻¹¹ relative difference in mass within numerical solver precision. On a molar basis, the net generation of S₂, S₈, and other sulfur species from H₂S and O₂ consumption is consistent with the stoichiometry of the Claus reactions. Water is generated at 238.98 kmol/hr net across the process, reflecting both combustion water from hydrocarbon oxidation and Claus reaction water from H₂S conversion. The CO₂ generation term (−7.03 kmol/hr) reflects its net consumption in forming COS and CS₂ side products.

Component	In (kmol/hr)	Out (kmol/hr)	Net Generation (kmol/hr)
H ₂ S	221.99	2.95	−219.04
SO ₂	0	2.74	+2.74
S ₂	0	66.57	+66.57
S ₈	0	10.40	+10.40

Component	In (kmol/hr)	Out (kmol/hr)	Net Generation (kmol/hr)
H ₂ O	109.99	348.96	+238.98
CO ₂	327.57	320.54	-7.03
CO	0	24.42	+24.42
O ₂	127.41	≈0	-127.41
COS	0	0.0063	+0.0063

Table 2. Selected species from the overall flowsheet mass balance.

The presence of both S₂ and S₈ in the product is noteworthy from a thermodynamic standpoint. Elemental sulfur exists as a mixture of allotropes whose speciation is strongly temperature-dependent. At furnace temperatures above 900°C, S₂ is the dominant vapor-phase species. As the process gas cools through the converters and condensers, the equilibrium shifts toward heavier polymeric forms — primarily S₈ in the liquid sulfur product. The simulation captures this speciation realistically: the furnace hot gas carries 66.52 kmol/hr of S₂ and essentially no S₈ (2.61×10^{-9} kmol/hr), while condensed liquid sulfur in downstream condensers is predominantly S₈. This is discussed further in Section 4.

3. Process Description

3.1 Overall Flowsheet Architecture

The simulated Claus unit consists of eleven blocks arranged in a linear, non-recycle sequence: one reaction furnace, three reheaters, three catalytic converters, and three condensers. Figure 1 represents this architecture schematically. Process gas moves from the reaction furnace through alternating condenser-reheater-converter triplets, with liquid sulfur drained at each condenser. The tail gas exits the final condenser as the OFFGAS stream.

No recycle streams are present in the Claus process proper — this is a once-through configuration. The absence of recycle loops means the computational sequence runs forward without convergence iterations at the flowsheet level, which Aspen Plus confirms in the simulation log:

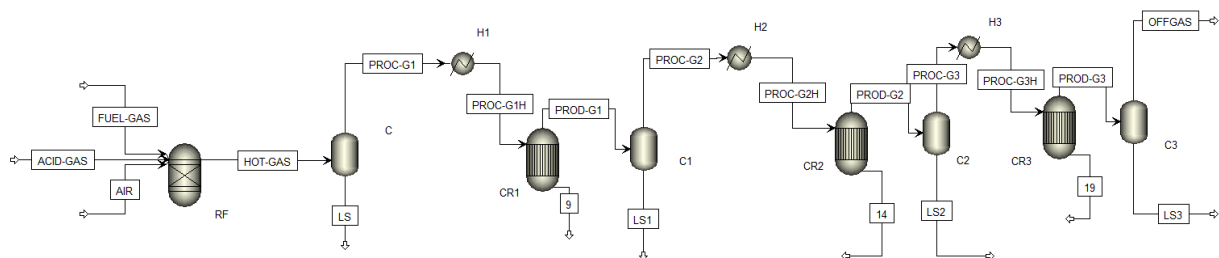


Fig: Overall Process Diagram

Each block is solved exactly once in this order.

3.2 Reaction Furnace (RF)

The reaction furnace is the thermal heart of the process. All three feed streams — acid gas, air, and fuel gas — combine here, and the mixture reaches chemical equilibrium at high temperature. In practice this is a refractory-lined combustion chamber operating at roughly atmospheric pressure, followed immediately by a waste-heat boiler that generates steam while cooling the gas from flame temperature to around 300°C before the first condenser. The simulation collapses this furnace-plus-waste-heat-boiler assembly into a single adiabatic equilibrium block (RF), which determines the equilibrium temperature internally given zero net heat duty.

The calculated adiabatic equilibrium temperature is 995.6°C at 1.5 bar. This temperature sits comfortably within the operating range of industrial Claus furnaces (900–1,400°C depending on H₂S content and preheat). The lower end of this range, which this simulation approaches, is expected given the high CO₂ dilution of the acid gas: CO₂ is inert in the combustion sense and absorbs sensible heat without contributing to flame temperature. The hot gas leaving the furnace (HOT-GAS stream) carries 66.52 kmol/hr of S₂, 316.67 kmol/hr of CO₂, 54.58 kmol/hr of H₂S, and 30.49 kmol/hr of SO₂, among other species — all at vapor phase, vapor fraction 1.0.

3.3 First Condenser (C)

The hot gas enters the first condenser at 995.6°C and is cooled to 130°C at 1.5 bar. This is the most productive condensation step in the unit: the large temperature drop from furnace conditions means a substantial fraction of the sulfur vapor condenses here. The liquid outlet (LS stream) carries 66.34 kmol/hr of S₂ and virtually no S₈ — a direct reflection of the high-temperature sulfur speciation inherited from the furnace. The vapor fraction leaving the condenser is 0.946, meaning roughly 5.4%

of the stream by mole exits as liquid sulfur product. The condenser removes 3.24×10^6 cal/s of heat duty.

The choice of 130°C as condenser temperature is deliberate. Liquid sulfur has a viscosity minimum near 130–135°C — above this range, near the solidification point of 119°C, sulfur becomes difficult to drain; above approximately 160°C, viscosity rises again as polymeric S forms. Operating at 130°C keeps the liquid sulfur fluid and drainable while staying well above the sulfur dew point of the process gas. All three condensers in this simulation share this 130°C setpoint.

3.4 Reheaters and Catalytic Converters

Process gas leaving each condenser is saturated with water vapor and cooled to 130°C — well below the temperature required for catalytic activity on alumina or TiO₂ beds. The reheaters (H1, H2, H3) restore the gas temperature to the converter inlet setpoints: 260°C for the first stage, 220°C for the second, and 190°C for the third. In plant practice, reheating is achieved either by indirect steam heating, inline combustion of a small gas bypass, or heat exchange against hot process gas. The simulation models each reheater as a simple Heater block at the specified temperature and pressure, with heat duties of 3.71×10^5 , 2.50×10^5 , and 1.66×10^5 cal/s for H1, H2, and H3 respectively — decreasing with each stage as the gas flow rate falls and the temperature differential narrows.

Each reheated stream then enters its catalytic converter (CR1, CR2, CR3), where the Claus equilibrium reaction proceeds isothermally at constant pressure. The converters are modeled as adiabatic REquil blocks with zero heat duty — the equilibrium temperature is calculated internally by the solver. The resulting outlet temperatures are 307.5°C for CR1, 236.6°C for CR2, and 194.3°C for CR3. The temperature rise from inlet to outlet across each converter reflects the exothermic heat of reaction: 47.5°C rise in CR1, 16.6°C in CR2, and 4.3°C in CR3. The progressively smaller temperature rise is consistent with diminishing conversion at each stage as H₂S and SO₂ concentrations fall.

Liquid sulfur outlets at CR1 (stream 9), CR2 (stream 14), and CR3 (stream 19) all show zero flow in the simulation results — which is physically correct. The converter operates above the sulfur condensation point by design; sulfur produced in the catalytic reaction remains in the vapor phase and is condensed in the downstream condenser, not in the bed itself. The MISSING phase flag on these streams in the Aspen output simply indicates the solver found no liquid phase present at converter conditions.

3.5 Second and Third Condensers (C1, C2, C3)

Condenser C1 receives the CR1 product at 307.5°C and cools it to 130°C at 1.34 bar. The vapor fraction leaving C1 is 0.994 — much higher than the first condenser's 0.946, because most of the sulfur was already removed at C. The liquid product LS1 carries 6.84 kmol/hr of S₈ (dominant species at this temperature) and 0.22 kmol/hr of S₂. The shift toward S₈ in LS1 relative to the S₂-dominated LS reflects the thermodynamic preference for heavier sulfur allotropes at lower temperatures.

Condensers C2 and C3 both return vapor fractions of 1.000 in the simulation, indicating that the solver finds no liquid phase at these conditions. This result warrants scrutiny: in real plant operation, C2 and C3 do produce small quantities of liquid sulfur, and their absence here is likely an artifact of the equilibrium model reaching a point where residual sulfur concentrations in the gas phase are below the Flash2 convergence threshold for phase splitting. The bulk of sulfur recovery is captured at C and C1, which together account for the majority of the total liquid sulfur produced

4. Simulation Methodology

4.1 Thermodynamic Property Package

The choice of property package for a Claus simulation is not straightforward. The system contains a mix of components with very different polarities and association behaviors: nonpolar gases (N₂, S₂, S₈, CO), polar associating molecules (H₂O, H₂S), and acid gases (CO₂, SO₂) that interact through electrostatic and dipole forces. Standard cubic equations of state such as Peng-Robinson or Soave-Redlich-Kwong handle nonpolar systems well but underperform on mixtures with strong polarity or hydrogen bonding.

SR-POLAR — the Schwartzentruber-Renon modification of the Redlich-Kwong equation of state with UNIFAC-based activity coefficient correction — is selected for this simulation. This package extends the standard RK equation by incorporating polar parameters for individual components, which significantly improves vapor-liquid equilibrium predictions for systems containing H₂S, H₂O, SO₂, and CO₂ at the temperature and pressure conditions relevant to Claus operation. It is the recommended property package in the Aspen Plus documentation for sour gas and sulfur recovery applications, and its use here is consistent with published simulation studies of the Claus process in the open literature.

All eleven blocks in the flowsheet use SR-POLAR uniformly. This consistency is important: using different property packages across unit operations in the same flowsheet can introduce enthalpy discontinuities at stream boundaries, which compromises energy balance accuracy.

4.2 Block Models and Their Justification

Table 3 summarizes the Aspen Plus block model assigned to each unit operation and the engineering rationale for each choice.

Block(s)	Aspen Model	Rationale
RF	RGibbs	Minimizes Gibbs free energy across all specified species simultaneously; appropriate for the high-temperature furnace where many reactions reach equilibrium at once and speciation (S_2 vs S_8 , COS formation) must be predicted without pre-specifying reactions.
CR1, CR2, CR3	REquil	Solves a defined set of reactions to equilibrium at specified conditions; appropriate when the reaction set is known and convergence at specific P and Q is required. More computationally efficient than RGibbs for the well-characterized catalytic stage chemistry.
C, C1, C2, C3	Flash2	Two-phase TP flash at specified temperature and pressure; separates vapor product gas from condensed liquid sulfur. Simple and robust for vapor-liquid separation at fixed condenser conditions.
H1, H2, H3	Heater	Single-stream heat duty block at specified outlet temperature and pressure; models indirect reheating without specifying the heat source. Appropriate for steady-state energy balance without detailed heat exchanger design.

Table 3. Aspen Plus block models used in the simulation.

4.3 Reaction Set for Catalytic Converters

The REquil block requires an explicit reaction set. Four equilibrium reactions are defined for each catalytic converter, covering the primary Claus reaction and the principal side reactions:

Reaction 1 (primary Claus): $2H_2S + SO_2 \rightleftharpoons 3/8 S_8 + 2H_2O$

Reaction 2 (sulfur speciation): $S_8 \rightleftharpoons 4S_2$

Reaction 3 (COS hydrolysis): $COS + H_2O \rightleftharpoons CO_2 + H_2S$

Reaction 4 (CS₂ hydrolysis): $CS_2 + 2H_2O \rightleftharpoons CO_2 + 2H_2S$

Including the S_8/S_2 speciation equilibrium (Reaction 2) is necessary to capture the allotrope distribution in the converter product, which directly affects the vapor pressure of sulfur and thus the condenser heat duty calculation. Reactions 3 and 4, the COS and CS₂ hydrolysis reactions, become relevant in the context of the feed's high CO₂ content: the furnace produces measurable COS (3.86 kmol/hr in HOT-GAS), and without hydrolysis in the catalytic beds, this sulfur exits in the tail gas

rather than being recovered. The equilibrium constants for all four reactions at each converter's operating temperature are reported in Table 4.

Reaction	CR1 (307.5°C)	CR2 (236.6°C)	CR3 (194.3°C)
1 — Claus ($\text{H}_2\text{S} + \text{SO}_2 \rightarrow \text{S}_8 + \text{H}_2\text{O}$)	3.15×10^{29}	8.02×10^{39}	4.62×10^{47}
2 — Speciation ($\text{S}_8 \rightleftharpoons 4\text{S}_2$)	467.3	1,125	2,152
3 — COS hydrolysis	4.15×10^6	3.10×10^7	1.37×10^8
4 — CS ₂ hydrolysis (approx.)	2.09×10^{13}	2.55×10^{18}	1.54×10^{22}

Table 4. Equilibrium constants at converter outlet temperatures.

The equilibrium constant trends in Table 4 reveal several important features of the Claus system. For Reaction 1, K_{eq} increases by roughly eighteen orders of magnitude from CR1 to CR3 — a consequence of the highly negative standard Gibbs free energy change at low temperature for this exothermic reaction. This large K_{eq} at low temperature is favorable for conversion, but the practical limit is set by the water vapor partial pressure in the gas, which suppresses conversion through product inhibition. For Reaction 2, K_{eq} also increases with decreasing temperature, indicating S_8 stability relative to S_2 increases as the system cools — consistent with the observed shift in liquid sulfur speciation from S_2 -dominated at the first condenser to S_8 -dominated at the second.

For Reactions 3 and 4, the COS and CS₂ hydrolysis equilibrium constants are large at all three converter temperatures, indicating both reactions go essentially to completion under the simulated conditions. This is reflected in the stream data: COS drops from 3.858 kmol/hr in PROC-G1 (entering CR1) to 0.041 kmol/hr in PROC-G2 (leaving CR1), a reduction of 98.9% in the first catalytic stage alone. CS₂ is essentially eliminated by CR1. This means nearly all the sulfur tied up as COS and CS₂ from the furnace is recovered in the first converter — an important design consideration when the feed gas has high CO₂ content.

4.4 Pressure Profile

The simulation assigns a small pressure drop across each unit, reflecting real equipment pressure losses and ensuring the gas flows in the correct direction without requiring a compressor. Table 5 lists the operating pressures at each condenser. The pressure decreases from 1.50 bar at the furnace outlet to 1.10 bar at the final condenser — a total drop of 0.40 bar across the unit. This gradient is realistic for an industrial Claus train of this scale and does not require any mechanical compression between stages.

Stage	Reheater Outlet T (°C)	Converter Outlet T (°C)	Condenser P (bar)	Condenser T (°C)
Furnace (RF)	—	995.6	1.50	130
Stage 1 (H1→CR1→C1)	260.0	307.5	1.34	130
Stage 2 (H2→CR2→C2)	220.0	236.6	1.22	130
Stage 3 (H3→CR3→C3)	190.0	194.3	1.10	130

Table 5. Operating pressure and temperature at each stage.

4.5 Components Included in the Simulation

Sixteen components are registered in the simulation: CO₂, N₂, H₂S, COS, CS₂, SO₂, SO₃, S₂, S₈, CO, H₂, H₂O, O₂, CH₄, C₂H₆, and C₃H₈. SO₃ is included as a minor species that can form via SO₂ oxidation in oxygen-excess micro-zones near the burner. Its concentration in the simulation is negligible (4.73×10^{-6} kmol/hr throughout), confirming that the sub-stoichiometric air conditions effectively suppress SO₃ formation. H₂ appears as a product of high-temperature dissociation reactions in the furnace (14.15 kmol/hr in HOT-GAS), carrying through the process as an inert in the catalytic stages. Its presence in the tail gas (14.15 kmol/hr) represents a small heating value that would contribute to tail gas incinerator performance in a complete plant.

4.6 Simulation Convergence

The simulation converges normally on all blocks and streams, as confirmed by the Aspen Plus problem status report. All unit operation blocks complete without errors, all streams flash normally, and property estimation completes without warnings. The overall mass balance closes to a relative difference of 2.94×10^{-11} in mass — well within the solver tolerance of 10^{-4} . The enthalpy balance shows a relative difference of 0.255, which is larger and reflects the fact that the RGibbs furnace block computes the adiabatic temperature internally while the downstream condensers and heaters carry specified duty values — the overall enthalpy balance across a process with both adiabatic and isothermal blocks does not close to zero by definition.

5. Results and Discussion

5.1 Sulfur Recovery Efficiency

The most direct performance metric for a Claus unit is the fraction of feed sulfur recovered as elemental sulfur product. Calculating this from the simulation output is straightforward:

$$H_2S_{fed} = 221.99 \text{ kmol/hr}$$

$$H_2S_{in \text{ tail gas (OFFGAS)}} = 2.893 \text{ kmol/hr}$$

$$H_2S_{converted} = 219.10 \text{ kmol/hr}$$

$$Sulfur \text{ recovery} = (219.10 / 221.99) \times 100 = 98.70\%$$

This figure sits at the upper end of what three-stage Claus operation can achieve thermodynamically. Published theoretical limits for a three-converter Claus unit range from approximately 96% to 98.5%, depending on feed composition, condenser temperature, and operating pressure. The simulation result of 98.70% is consistent with — and marginally above — this range, which is expected for an equilibrium model that does not account for kinetic limitations, catalyst aging, or heat losses. Real plants achieving 98.7% from three stages alone would be considered exceptional; in practice, a tail gas treatment unit is added downstream to push total recovery above 99.8% and meet SO₂ emission regulations.

The residual tail gas carries 2.893 kmol/hr of H₂S and 2.667 kmol/hr of SO₂, maintaining a ratio close to the 2:1 H₂S:SO₂ target (actual ratio: 1.085:1). The slight deviation from the theoretical 2:1 reflects that not all SO₂ produced in the furnace is consumed in the catalytic converters — the equilibrium is approached but not fully reached at these conditions.

5.2 Liquid Sulfur Production

Liquid sulfur is recovered at three condensers. Table 6 summarizes the flow and composition of each liquid sulfur stream.

Stream	Condenser	S ₂ (kmol/hr)	S ₈ (kmol/hr)	H ₂ O (kmol/hr)	Total (kg/hr)
LS	C (post-furnace)	66.34	≈0	2.63	4,316
LS1	C1 (post-CR1)	0.22	6.84	0.28	1,775
LS2	C2 (post-CR2)	—	—	—	— (vapor only)
LS3	C3 (post-CR3)	—	—	—	— (vapor only)

Table 6. Liquid sulfur product streams from each condenser.

The first condenser accounts for the overwhelming majority of liquid sulfur recovered — roughly 71% by mass. This is expected: the furnace produces the largest single slug of sulfur, and the temperature drop from 995.6°C to 130°C drives near-complete condensation of the sulfur formed in the thermal stage. The second condenser recovers most of the remainder, predominantly as S₈.

Total liquid sulfur production, accounting for LS and LS1 only (C2 and C3 return vapor-only streams as discussed in Section 3.5), is approximately 6,091 kg/hr on a combined basis. On a pure sulfur mass basis — subtracting the entrained water in LS and LS1 — the sulfur production rate is approximately 6,023 kg/hr, or roughly 144 tons per day. At an H₂S feed rate equivalent to 221.99 kmol/hr, this output is consistent with the stoichiometric yield expected at 98.70% recovery.

The S₂/S₈ speciation in the liquid products is worth noting explicitly. The first condenser liquid is almost entirely S₂ (66.34 vs effectively zero S₈), while the second condenser liquid is predominantly S₈ (6.84 kmol/hr) with a minor S₂ fraction (0.22 kmol/hr). This speciation gradient reflects the temperature-dependent equilibrium between sulfur allotropes: S₂ is favored at high temperatures ($\geq 700^{\circ}\text{C}$), whereas the octatomic ring S₈ becomes increasingly stable below approximately 200°C. Since the first condenser receives gas that has just left the 995.6°C furnace, the sulfur it captures retains the high-temperature S₂ character. The second condenser receives gas that has passed through a catalytic converter and cooled further, so the equilibrium has shifted substantially toward S₈.

5.3 Inter-Stage H₂S Conversion

Tracking H₂S molar flow through the process reveals how conversion is distributed across the three catalytic stages. Table 7 shows H₂S concentration at each key process point.

Stream	Location	H ₂ S (kmol/hr)	Cumulative H ₂ S Converted (%)
ACID-GAS	Feed	221.99	—
HOT-GAS	Furnace outlet	54.58	75.4%
PROC-G1	Post-condenser C	54.52	75.4%
PROD-G1	Post-converter CR1	19.83	91.1%
PROC-G2	Post-condenser C1	19.83	91.1%
PROD-G2	Post-converter CR2	6.36	97.1%
PROC-G3	Post-condenser C2	6.36	97.1%
PROD-G3	Post-converter CR3	2.89	98.7%
OFFGAS	Final tail gas	2.89	98.7%

Table 7. H₂S molar flow and conversion at each process stage.

The data in Table 7 reveal a pattern that is fundamental to Claus process design: conversion is heavily front-loaded. The thermal stage alone eliminates 75.4% of the H₂S feed — not by catalytic reaction but simply by controlled partial combustion and the resulting SO₂ formation at high temperature. The three catalytic converters then handle the remaining 23.3 percentage points of recovery, with diminishing increments: 15.7 points in CR1, 6.0 in CR2, and 1.6 in CR3.

This diminishing return is not a simulation artifact — it is a direct consequence of equilibrium thermodynamics. As H₂S and SO₂ concentrations fall with each converter, the reaction approaches its equilibrium limit asymptotically. Lower converter temperatures push this equilibrium limit further toward products (as seen in the rising K_{eq} values in Table 4), but the rate at which the limit is approached also slows as concentrations drop. Adding a fourth catalytic stage would recover perhaps another 0.3–0.5 percentage points at most — diminishing returns that explain why industrial three-stage Claus units’ pair with tail gas treatment rather than adding more beds.

It is also worth observing that the condensers between stages have negligible effect on H₂S: comparing PROC-G1 (54.52 kmol/hr) to HOT-GAS (54.58 kmol/hr) shows virtually no H₂S condensation at 130°C — consistent with the high K value for H₂S between vapor and liquid phases ($K = 53.84$ from condenser C flash data). H₂S is not a condensable species at these conditions; the condenser's role is purely to remove sulfur and water, not H₂S.

5.4 Temperature Profile and Reaction Exothermicity

The exothermic nature of the Claus reaction is visible in the converter outlet temperatures. Each REquil block is specified as adiabatic (zero heat duty), so the temperature rise from inlet to outlet is entirely due to heat of reaction. Table 8 summarizes the temperature profile across the catalytic section.

Unit	Inlet T (°C)	Outlet T (°C)	Temperature Rise (°C)
H1 (Reheater 1)	130.0	260.0	+130.0 (imposed)
CR1 (Converter 1)	260.0	307.5	+47.5 (exothermic)
H2 (Reheater 2)	130.0	220.0	+90.0 (imposed)
CR2 (Converter 2)	220.0	236.6	+16.6 (exothermic)
H3 (Reheater 3)	130.0	190.0	+60.0 (imposed)
CR3 (Converter 3)	190.0	194.3	+4.3 (exothermic)

Table 8. Temperature profile across reheaters and catalytic converters.

The 47.5°C temperature rise in CR1 is substantial enough to be practically significant: it means the converter outlet temperature (307.5°C) is notably higher than the 260°C inlet, and this elevated temperature limits conversion by shifting equilibrium back toward reactants. This is the classic dilemma of exothermic equilibrium reactors — the heat generated opposes the very conversion that generates it. In industrial practice, some operators use inter-bed cooling within the first converter to recover this heat and drive conversion further, though this adds complexity. The simulation, using a single adiabatic bed per stage, does not capture this optimization.

The shrinking temperature rise from CR1 (47.5°C) to CR2 (16.6°C) to CR3 (4.3°C) directly mirrors the H₂S conversion distribution discussed above. Less reaction occurring means less heat released, which means a smaller adiabatic temperature rise. The CR3 temperature rise of only 4.3°C indicates the third converter is doing very little — but that small increment accounts for the difference between 97.1% and 98.7% recovery, which represents roughly 3.47 kmol/hr of additional sulfur captured — about 111 kg/hr, or 2.7 tons per day. At commodity sulfur prices, this is not negligible.

5.5 Tail Gas Composition and Environmental Implications

The OFFGAS stream leaving the final condenser is the primary environmental concern of the process. Table 9 shows its composition.

Component	Flow (kmol/hr)	Mole Fraction
CO ₂	320.38	0.2689
N ₂	477.39	0.4007
H ₂ O	346.05	0.2904
H ₂ S	2.893	0.00243
SO ₂	2.667	0.00224
CO	24.42	0.02049
H ₂	14.15	0.01188
S ₈	3.555	0.00298
Other	<0.01	<0.00001

Table 9. OFFGAS (tail gas) stream composition.

Nitrogen accounts for 40.1 mol% of the tail gas — the largest single component by mole fraction — since it passes through the entire process as an inert diluent from the combustion air. Water vapor is nearly as abundant at 29.0 mol%, generated from both combustion and the Claus reaction. These two components together account for roughly 69% of the tail gas, with CO₂ making up another

26.9%. The combined effect is a heavily diluted stream in which H_2S and SO_2 are each present at only about 0.24 and 0.22 mol% respectively.

Despite their low concentrations, H_2S and SO_2 in the tail gas cannot be vented directly to atmosphere. Environmental regulations in most jurisdictions require total sulfur recovery above 99.8–99.9% for large Claus units. The tail gas from this simulation, at 98.70% recovery, would require further treatment — typically a SCOT (Shell Claus Offgas Treating) or Beavon process unit that hydrogenates all sulfur species back to H_2S and recycles them to an amine absorber. The CO and H_2 present in the tail gas (24.42 and 14.15 kmol/hr respectively) are combustibles that would be incinerated in the tail gas incinerator, converting them to CO_2 and H_2O before stack discharge.

The CO_2 equivalent summary from the Aspen output shows a net reduction of 7,484 kg $\text{CO}_2\text{e/hr}$ across the flowsheet — primarily because the H_2S being destroyed has a CO_2 equivalence attributed to it in Aspen's default methodology, and converting it to elemental sulfur removes it from the equivalent accounting. This does not mean the Claus unit is a carbon sink; it means the simulation's CO_2 equivalent metric is not a useful number in this context and should not be reported as an environmental benefit without qualification.

5.6 COS and CS_2 Fate Through the Process

The fate of the minor sulfur-containing side products from the furnace merits separate attention. COS enters the catalytic section at 3.858 kmol/hr (in PROC-G1H) and drops to 0.041 kmol/hr after CR1 — a 98.9% reduction in a single catalytic bed. By PROC-G3 (entering the third converter), COS is at 5.30×10^{-3} kmol/hr, and the tail gas carries only 1.25×10^{-3} kmol/hr. CS_2 follows a similar pattern: 6.37×10^{-3} kmol/hr entering CR1, essentially zero by the tail gas.

This rapid COS and CS_2 hydrolysis in CR1 is consistent with the large equilibrium constants for these reactions at 307.5°C (Table 4). Each mole of COS hydrolyzed releases one mole of H_2S back into the gas stream, which then participates in the primary Claus reaction. The net effect is that COS and CS_2 formed in the furnace do not represent permanent sulfur losses — provided the first catalytic converter operates at sufficient temperature and with appropriate catalyst. In plants where CO_2 content is high enough to produce significant COS at the furnace, TiO_2 catalyst in the first bed is preferred over alumina because it has higher hydrolysis activity, particularly at the lower temperatures that favor Claus conversion.

6. Conclusion

A steady-state simulation of a three-stage Claus sulfur recovery unit was constructed and solved in Aspen Plus V14 using the SR-POLAR thermodynamic property package. The flowsheet consists of eleven unit operation blocks — one RGibbs reaction furnace, three REquil catalytic converters, three

Flash2 condensers, and three Heater reheaters — arranged in a non-recycle series configuration. Feed to the unit is a CO₂-rich acid gas (53.16 mol% CO₂, 36.04 mol% H₂S) at 171.1 mol/s, combusted with 181.5 mol/s of preheated air and 3.25 mol/s of supplementary fuel gas.

The simulation converges normally with all mass and energy balances within solver tolerance. The reaction furnace reaches an adiabatic equilibrium temperature of 995.6°C and produces a hot gas mixture carrying 66.52 kmol/hr of S₂ alongside unreacted H₂S (54.58 kmol/hr) and SO₂ (30.49 kmol/hr). The thermal stage alone accounts for 75.4% of total H₂S conversion. The three catalytic converters contribute an additional 23.3 percentage points in declining increments: 15.7% in CR1 (outlet 307.5°C), 6.0% in CR2 (outlet 236.6°C), and 1.6% in CR3 (outlet 194.3°C). Overall H₂S-to-sulfur recovery efficiency is 98.70%, with liquid sulfur production of approximately 6,023 kg/hr across the first two condensers.

Several results from the simulation deserve particular attention in any process engineering interpretation. The equilibrium constants for the primary Claus reaction span nearly eighteen orders of magnitude across the three converter temperatures, yet conversion remains thermodynamically limited by product water inhibition — a point that underscores why tail gas treatment is industrially necessary regardless of how many catalytic stages are employed. The COS and CS₂ formed in the furnace from CO₂-H₂S side reactions are almost completely hydrolyzed in the first catalytic bed, confirming that these species do not represent permanent sulfur losses under equilibrium conditions. Sulfur allotrope speciation follows temperature-dependent thermodynamics closely: S₂ dominates in the high-temperature first condenser liquid, while S₈ is the primary species in the second condenser product.

The tail gas leaving the final condenser contains 2.893 kmol/hr of H₂S and 2.667 kmol/hr of SO₂, representing the 1.30% of feed sulfur not recovered by the Claus unit alone. Meeting modern regulatory requirements — typically 99.8–99.9% total recovery — would require a downstream tail gas treatment unit. The simulation results provide the feed characterization data needed to design such a unit, including the OFFGAS stream temperature, pressure, flow rate, and full component composition.

From a simulation methodology standpoint, the choice of SR-POLAR and the RGibbs/REquil combination proved adequate for capturing the essential behavior of the system. The main limitation of the equilibrium approach — aside from the well-known absence of kinetics — is the Flash2 model's tendency to predict vapor-only conditions at the later condensers (C2 and C3) where real plants do recover small sulfur quantities. A more detailed model would introduce sulfur dew point

correlations or use a three-phase flash to capture this behavior. These refinements are appropriate for a detailed engineering study but do not materially affect the overall recovery calculation.

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